AnaCraft: Duel-Play Probabilistic-Model-Based Reinforcement Learning for Sample-Efficient PVT-Robust Analog Circuit Sizing Optimization

Mohsen Ahmadzadeh, Jan Lappas, *Member, IEEE*, Norbert Wehn, *Senior Member, IEEE*, and Georges Gielen, *Fellow, IEEE*

***Abstract*—Recent advancements in machine learning offer the potential for finding faster and robust optimization approaches for analog circuit design automation. However, fully automated yet fast and process, voltage, and temperature (PVT)-robust sizing algorithms are still lacking as even the most recent methods continue to require extensive simulations or domain-specific circuit expertise. In this article, we present a PVT-robust analog circuit sizing method, called AnaCraft, that is the first to introduce an adversarial training scheme of multiagent reinforcement learning (RL) for robust circuit design automation. We adopt the soft actor–critic (SAC) agent for circuit sizing, which outperforms other actor–critic agents in stability and robustness. Then, we introduce a duel-play scheme to address PVT-robustness, where sizing agents cooperate to find optimal circuit parameters while competing with an adversarial PVT agent. We combine this approach with the model-based policy optimization method: an ensemble of probabilistic models is trained and used to extract many short rollouts of generated data for updating the sizing agents. We test our algorithm on the sizing of operational amplifiers in a 45-nm CMOS technology, as well as on a complex data receiver circuit in a predictive 7-nm FinFET technology. This demonstrates our approach's ability to find PVT-robust power-area-optimal sizes for advanced technologies and circuits. Our proposed method achieves a higher figure of merit with up to ∼3× fewer circuit simulations and ∼2× less runtime compared to existing state-of-the-art methods.**

***Index Terms*—Adversarial reinforcement learning (RL), circuit design automation, model-based policy optimization (MBPO), process, voltage, and temperature (PVT)-robustness, soft actor-critic (SAC).**

I. Introduction

DESPITE occupying only a small fraction of electronic chips, analog circuits still disproportionately require a large manual design effort due to the lack of widely used design automation tools [2]. Analog design usually involves finding circuit parameters that optimize the power and area and that satisfy many conflicting (hard) specification constraints while being robust against process, voltage and temperature (PVT) variations. The curse of dimensionality, however, continues to hinder also the most recent efforts for the fast optimization of analog circuits [1].

During the past decade, evolutionary algorithms (EA) [3], [4] and Bayesian optimization (BO) methods [5], [6] were the dominant simulation-based methods used in analog circuit sizing optimization. However, the former have limited learning capabilities due to their intrinsic randomness, resulting in a huge number of simulations needed to optimize complex circuits [7], while the latter suffer from cubic computational scaling in their Gaussian processes (GPs), which also makes them runtime expensive [8] and thus limited to simple circuits and problems that have less than 20 parameters to optimize [9]. Therefore, the recent trend in analog circuit design automation is to use reinforcement learning (RL) or RL-inspired algorithms that find solutions with both a better performance and in fewer simulations compared to EA and BO [7], [10], [11], [12], [13], [14], [15], [16], [22], [23].

Most existing RL-based sizing methods utilize the deep deterministic policy gradients (DDPG) algorithm [10], [11], [12], [13], [14]. Recently, the Twin-Delayed DDPG (TD3) method has been proposed to improve the robustness across different runs; it can also be used in a multiagent RL (MARL) scheme for sizing complex circuits [1], [16]. Both DDPG and TD3, however, require significant hyperparameter tuning and still exhibit considerable variance and fluctuations across different runs. Moreover, many of these methods focus only on nominal conditions and ignore PVT variations, which can cause significant performance degradation in analog circuits [18], [19], [20].

To address PVT robustness, a few prior works have introduced techniques, such as a combination of K-means clustering with multitask RL in *RobustAnalog* [13], or have embedded BO in their RL framework, such as *Rose* [21] and *PVTSizing* [19]. These methods, however, either have limited PVT inference capabilities and suffer from sample inefficiency, or are challenging to integrate into a MARL framework for complex analog circuits. Existing PVT-robust sizing methods primarily aim to find sizing parameters that satisfy the hard constraints across PVT-variability, without further optimizing for power and area, falling short of achieving truly optimal and robust designs under PVT variations.

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Mohsen Ahmadzadeh and Georges Gielen are with the Department of Electrical Engineering, KU Leuven, 3001 Leuven, Belgium (e-mail: mohsen.ahmadzadeh@kuleuven.be; georges.gielen@kuleuven.be).

Jan Lappas and Norbert Wehn are with the Microelectronic Systems Design Research Group, University of Kaiserslautern–Landau, 67663 Kaiserslautern, Germany.

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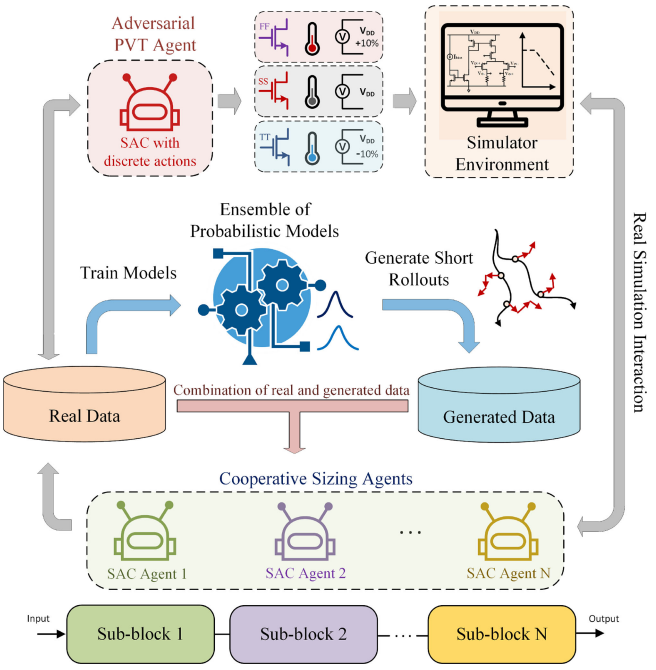


Fig. 1. Overview diagram of the AnaCraft duel-play MBPO approach for analog circuit sizing.

In this article, we introduce AnaCraft, which is a probabilistic-model-based RL algorithm for PVT-robust and sample-efficient analog circuit sizing automation using an adversarial PVT agent dueling with cooperative sizing optimization agents. It is a novel approach that addresses the above limitations by employing a competitive PVT agent and probabilistic models for the cooperative sizing agents. In this adversarial RL training method, which we refer to as a *duel-play RL* scheme, the sizing agents of the different circuit blocks cooperate to optimize the circuit performance while competing with a PVT enemy agent that suggests challenging corners to degrade the performance. Additionally, we leverage model-based RL techniques to accelerate the design exploration using dynamically updated local models, reducing the extensive need for actual circuit simulations. Our approach ensures PVT-robustness while exploring for power-area optimal solutions, demonstrating significant improvements in both efficiency and robustness of the analog circuit sizing task. Fig. 1 gives a general overview of our duel-play multiagent model-based RL scheme for circuit sizing.

The main contributions and features of this AnaCraft paper are the following.

1) Integration of the more robust soft actor–critic (SAC) agent into our RL sizing framework.
2) Development of the duel-play RL scheme where sizing agents compete with an adversarial PVT agent to find PVT-robust solutions that also optimize the power, area, and/or other metrics while satisfying multiple hard constraint specifications.
3) Implementation of probabilistic models to create a model-based policy optimization (MBPO) algorithm, further enhancing the sample efficiency of our duel-play RL algorithm.
4) Demonstration of the sizing optimization for two operational amplifiers with full PVT-robustness in a predictive 45-nm CMOS technology, showcasing the efficacy and efficiency of our duel-play MBPO method.
5) Extension of the algorithm to incorporate multiple cooperative sizing agents for complex circuits that are comprised of multiple sub-blocks. This is illustrated through the sizing optimization of a real-world high-speed data receiver circuit in a predictive 7-nm FinFET technology. This further demonstrates how cooperative sizing agents can compete against an adversarial PVT agent to ensure PVT-robustness for complex circuits.

The remainder of this article is organized as follows. Section II briefly reviews some background works. Then, in Section III, we explain the details of the AnaCraft approach: the SAC agent, the duel-play RL scheme, the probabilistic models, and the final multiagent MBPO setup. In Section IV, we present the experimental results of the duel-play multiagent MBPO for two opamps and a data receiver circuit. Finally, we conclude this article in Section V.

II. Background

The following background information is needed for the work presented in this article.

The *SAC* agent [17] demonstrates a greater robustness and reduced hyperparameter sensitivity compared to vanilla DDPG and TD3 agents, making it more suitable for more sophisticated model-based RL frameworks for analog circuit sizing. SAC achieves this by incorporating entropy maximization into its policy optimization, encouraging exploration and stabilizing learning by avoiding premature convergence to suboptimal policies [17].

Adversarial self-play RL is a powerful technique where agents learn by competing against each other. This approach has gained prominence in training algorithms for complex games like chess, Go, and StarCraft II, where agents co-evolve and refine their strategies through cooperative and adversarial interactions [24], [25], [26]. Inspired by these methods, we introduce our duel-play RL approach for more automated robust circuit sizing.

Model-based RL methods accelerate design exploration by using dynamically updated local models to generate short predictive rollouts, reducing the need for actual simulations [27], [28]. Model inaccuracies, however, can introduce destructive bias into the policy. MBPO [28] addresses this by training ensembles of probabilistic dynamic models to account for uncertainties and using synthetic rollouts alongside real data to update the RL agent's neural networks. We demonstrate that MBPO enables efficient and effective analog circuit sizing.

III. Duel-Play Model-Based RL

Formally, the PVT-robust analog circuit sizing optimization problem addressed in this work can be stated as a constrained multiobjective optimization problem

$$\begin{aligned} \underset{\mathbf{x}}{\text{minimize}} \quad & \mathbf{f}_{\text{obj}}(\mathbf{x}) = \left[f_1(\mathbf{x}), \ldots, f_k(\mathbf{x})\right]^T \\ \text{subject to} \quad & g_j(\mathbf{x}, c) \ \leq Y_j \end{aligned} \tag{1}$$

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$$\begin{aligned} &\forall j = 1, \ldots, m_g \quad \forall c \in C \\ h_l(\mathbf{x}, c) \ &\geq Y_l \\ &\forall l = 1, \ldots, m_h \quad \forall c \in C \\ \mathbf{x} &\in D. \end{aligned}$$

Here, $\mathbf{x} \in \mathbb{R}^d$ is the vector of d circuit design parameters within the allowed design space D. The goal is to minimize a vector of T optimization targets $\mathbf{f}_{\text{obj}}(\mathbf{x})$, while satisfying $m = m_g + m_h$ performance constraints. These constraints, denoted by $g_j(\mathbf{x}, c) \leq Y_j$ (upper-bound constraint) and $h_l(\mathbf{x}, c) \geq Y_l$ (lower-bound constraint), must hold for the design $\mathbf{x}$ across all specified PVT corner conditions c within the discrete set C. The core challenge lies in finding a parameter vector $\mathbf{x}$ that not only optimizes the objectives $\mathbf{f}_{\text{obj}}(\mathbf{x})$ but that also crucially satisfies all performance constraints (g_j and h_l) across the *entire* set of PVT corners C, within the design parameters' search space D. Efficient approaches are needed to search this complex, high-dimensional and constrained space.

A. Reinforcement Learning Setup for Analog Circuit Sizing

Formulating any problem in an RL framework requires three main elements: 1) the reward definition; 2) the state space; and 3) the action space. In this section, we define these elements for the analog circuit sizing optimization problem.

Reward: The Figure-of-Merit (FoM) used as the reward signal is comprised of reward terms for both hard constrained performance specifications as well as optimization targets. Each hard constrained specification can be in one of two sets: 1) the set Y_U includes specifications with an upper bound constraint, meaning that the specification should be lower than the constraint and 2) the set Y_L includes specifications with a lower bound constraint, meaning that the specification should be higher than the constraint value. Depending on this, the reward term for each hard constrained specification y is

$$z_y = \frac{y - y^*}{y + y^*} \quad \rightarrow \quad r_y = \begin{cases} \min(z_y, 0), & y \in Y_L \\ -\max(z_y, 0), & y \in Y_U \end{cases} \tag{2}$$

where y is the current specification value, y^* is the constraint value, and z_y is the normalized value of the performance relative to the constraint. The min and $-$max functions make sure that each reward term is always negative when the performance has not reached the constraint value, or zero when the constraint is satisfied. It should be noted that all performance values are bound to positive values in our code and therefore y is always made to be positive. This prevents y from having the opposite sign of y^*. Therefore, if for example the measured phase margin happens to be a negative value, we clip it to 0, which is the worst performance possible in our setup. Once we have all these r_y terms, the total reward term of the hard constrained specifications is

$$r_H = \sum_{y \in Y} r_y. \tag{3}$$

For any optimization target t (e.g., power and area), on the other hand, we do not apply the max and min functions to let these performances further influence the final reward value even after some target value (t^*) is met

$$o_t = \frac{t - t^*}{t + t^*} \quad \rightarrow \quad r_t = -o_t \quad \rightarrow \quad r_T = \sum_{t \in T} r_t \tag{4}$$

where o_t is the normalized term of the optimization target t. The final reward (FoM), therefore, is a weighted sum of the reward terms for the hard constrained specifications and the optimization targets from (3) and (4)

$$\text{FoM} = \begin{cases} r_H + \alpha \times r_T, & \text{if } r_H < -\epsilon \\ 0.3 + \beta \times r_T, & \text{if } r_H \geq -\epsilon \end{cases} \tag{5}$$

where ϵ can be zero or a small value like 0.02 depending on the level of tolerance we allow for the hard constraints. The above equation indicates that when not all hard constraints are satisfied ($r_H < -\epsilon$), the reward is comprised of both r_H and the weighted r_T ($\alpha = 0.05$ in our experiments) to train the agent on all performance metrics. When all hard constraints are satisfied [$r_H \geq -\epsilon$ according to (3)], a bonus positive reward of 0.3 is used as an indication of success; this bonus reward is added to r_T with factor β ($\beta = 1.0$ for current and area in our experiments) to push the agent toward further improving the optimization targets while keeping the constraints satisfied.

State Space: The state of the circuit at each step consists of the list of the normalized values of all performances (both hard constraints and optimization targets)

$$s = [z_1, \ldots, z_Y, o_1, \ldots, o_T]. \tag{6}$$

This state representation, based on normalized performance outcomes, provides sufficient information for the actor policy $\pi(a|s)$ to learn effective design parameters a that optimize the FoM, while maintaining scalability for complex circuits. Although the subsequent state s' resulting from simulating the action is primarily determined by a and the selected PVT corners (rather than s), this adheres to the Markov Property, as the transition remains dependent only on the current state-action pair (here, only the current action) and not on prior history. This formulation is common in RL-based optimization and aligns well with the predictive task of the probabilistic models introduced in Section III-D below.

Action Space: The action at each step consists of assigning a value to all the circuit design parameters that need to be sized. These involve the width (W), length (L) and multiplier (m) for each transistor; as well as the resistance values (R), the capacitance values (C), and any biasing voltages (V_i) in the circuit

$$a = [W_1, L_1, m_1, \ldots, W_n, L_n, m_n, R_1, \ldots, C_1, \ldots, V_1, \ldots]. \tag{7}$$

The predicted actions by the RL agent are in the range of $[-1, 1]$ and must be refined to the actual range of the design parameters for all simulations. For the FinFET technology, instead of W_i and m_i, the action space involves the number of fins (N_i^{fin}) and the number of fingers (N_i^f) for each transistor. It should be noted that the multiplier parameter (m_i) creates multiple parallel instances of a device, while the fingers

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Algorithm 1 Model-Free SAC for Circuit Sizing

1: **Input:** θ_1, θ_2, ϕ {Initialize Critics and Actor parameters}
2: $\bar{\theta}i \leftarrow \theta_i$ for $i \in 1, 2$ {Initialize target critic weights}
3: $\mathcal{D} \leftarrow \emptyset$ {Initialize an empty replay buffer}
4: **for** each episode **do**
5: **for** each environment step **do**
6: $a \sim \pi\phi(a|s)$ {Sample action from the Actor policy}
7: $s' \sim$ Simulator(a) {Simulate and obtain next state}
8: $r = FoM$ {Calculate the reward from eq. (2)-(5)}
9: $\mathcal{D} \leftarrow \mathcal{D} \cup (s, a, r, s')$ {Store in replay buffer}
10: **end for**
11: **if** step $>$ W warmup steps **then**
12: Sample mini-batch from $\mathcal{D}$
13: $\theta_i \leftarrow \theta_i - \lambda_Q \nabla_{\theta_i} J_{Q_i}(\theta_i)$ {Update the twin Critics}
14: $\phi \leftarrow \phi - \lambda_\pi \nabla_\phi J_\pi(\phi)$ {Update Actor weights}
15: $\bar{\theta}_i \leftarrow \tau\theta_i + (1-\tau)\bar{\theta}_i$ {Update target critics}
16: **end if**
17: $s \leftarrow s'$ {Set the state to the next state}
18: **end for**
19: **Output:** Optimized circuit sizing parameters

parameter (N_i^f) divides a single transistor's width into multiple parallel sections. The fin parameter (N_i^{fin}) specifies the number of thin conductive channels that form the FinFET transistor's gate.

B. Soft Actor–Critic Agent

Deep RL agents often struggle with requiring many simulations and having unstable convergence, necessitating a precise hyperparameter tuning. Vanilla Deep Deterministic Policy Gradient (DDPG) agents are particularly sensitive to the hyperparameters and suffer from brittle convergence [17]. The Twin-Delayed DDPG (TD3) method improves the robustness and is effective for analog circuit sizing [1], [16], but its learning curve and performance remain sensitive to the hyperparameters, especially if one attempts to equip the agents with models that typically bring in even more hyperparameters.

The SAC agent has been introduced to address these issues by incorporating entropy regularization into the actor–critic framework [17]. The entropy, in this context, measures the randomness or uncertainty of the agent's policy, encouraging exploration. For discrete actions, the entropy is defined as $H(\pi(\cdot|s)) = -\Sigma\pi(a|s)\log(\pi(a|s))$, where $\pi(a|s)$ is the probability of taking action a in state s. For continuous actions, it is the differential entropy: $h(\pi(\cdot|s)) = -\int \pi(a|s)\log(\pi(a|s))da$, where $\pi(a|s)$ is the probability density of taking action a in state s, and the integral is over all possible actions.

In actor–critic RL agents, the goal of the critic is to output a good approximation of the Q-Function for action a taken in state s (which is the expected value of cumulative future rewards until the end of the trajectory), while the goal of the actor is to suggest an action a in state s that maximizes that Q-Function. Algorithm 1 shows the model-free SAC agent for analog circuit sizing. It optimizes two loss functions: the soft Q-function loss, minimizing the Bellman residual

$$J_{Q_i}(\theta_i) = \mathbb{E}_{(s,a)\sim\mathcal{D}}\left[\left(Q^i_{\theta_i}(s,a) - \text{Target}\right)^2\right]$$
$$\text{Target} = r + \gamma\left(\min_i Q^i_{\bar{\theta}_i}(s',a') - \alpha\log\pi(a'|s')\right) \quad (8)$$

and the policy loss, maximizing the reward-entropy tradeoff

$$J_\pi(\phi) = \mathbb{E}_{(s,a)\sim\mathcal{D}}\left[\min_i Q^i_{\theta_i}(s,a) - \alpha\log\pi_\phi(a|s)\right] \quad (9)$$

where Q_i is the ith critic network (with parameters θ_i), the target critic $Q^i_{\bar{\theta}_i}$ is a delayed, periodically updated copy of the critic Q_i (line 15 in Algorithm 1) that provides a more stable and consistent target for the Q-function updates, reducing the variance during training. π_ϕ represents the policy (actor network with parameters ϕ), r is the reward of the current step and s' and a' are the state and action of the next step. The entropy is incorporated into the loss functions through the term $\alpha\log\pi$, where α serves as a parameter that controls the strength of the entropy regularization. It is the main hyperparameter to be tuned for effectively modulating the exploration-exploitation tradeoff. In this way, the SAC agent avoids getting stuck in local optima, becomes more robust to changes in the environment and hyperparameters, shows less fluctuations across different runs, and finds better solutions through improved learning. In our experiments, $\alpha = 0.01$ yielded a robust performance, enabling SAC to outperform TD3 in speed and maximum Figure of Merit (FoM). It should be noted that, similar to TD3, SAC employs twin critic networks to mitigate the maximization bias of vanilla DDPG.

C. Duel-Play Competitive and Cooperative Agents

MARL is usually used for tasks in which multiple agents require a complex coordination to cooperate or compete in order to accomplish difficult goals. Cooperative MARL has been shown to have a superior performance with respect to single-agent RL in circuit sizing [1], [16]. In these previous works, for sizing a complex analog circuit comprised of multiple sub-blocks, one separate agent was assigned to the sizing task of each sub-block. However, these works did not consider sizing for PVT robustness.

Since the performances of the circuit for each set of design parameters can be different in different PVT corners, we propose a competitive or adversarial PVT agent for proposing alternative corners at each step. The addition of this new adversarial agent creates our duel-play RL scheme. Algorithm 2 and Fig. 2 describe the overall duel-play procedure. We will explain these two types of agents now.

1) Adversarial PVT Agent: In our duel-play scheme, the PVT agent is an extra SAC agent with the explicit goal of degrading the performance of the circuit by suggesting a more difficult corner at each step. Here, we first explain its reward, state space, and action space; which are obviously different from those of the sizing agents explained in section A above.

PVT Reward: At each step of our duel-play scheme, the circuit is simulated under two corners with the set of parameters (action) suggested by the sizing actor. First, we randomly choose a corner, simulate the circuit, and obtain the reward r_1

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Algorithm 2 Duel-Play Model-Free Multiagent SAC for PVT-Robust Analog Circuit Sizing

Initialize SAC for sizing agents and the PVT agent
Initialize replay buffer $\mathcal{D} \leftarrow \emptyset$
for each episode **do**
  for each step **do**
    $(s, a, r, s') \sim$ `Two_Corner_Simulation` (s)
    Store transitions (s, a, r, s') of all agents in $\mathcal{D}$
    if step $> W$ warmup steps **then**
      Sample mini-batch from $\mathcal{D}$
      Update the actor–critic networks of all agents
    end if
    $s \leftarrow s'$ {Set the states to the next states}
  end for
end for

Function `Two_Corner_Simulation` (*state* s)
  Sample action set a_{size} from sizing agent actors
  Select random PVT corner c_1
  Simulate circuits with a_{size} in the c_1 corner;
  $\rightarrow$ Obtain reward r_1 and next sate s'_1
  $c_2 \sim a_{pvt}$ {Sample corner c_2 from PVT agent actor}
  Simulate circuits again with a_{size} but in the c_2 corner
  $\rightarrow$ Obtain reward r_2 and next state s'_2
  for each sizing agent i **do**
    Compute next state $s^{i'}$ and reward R^i
  end for
  Compute PVT agent reward R_{pvt} using eq. (10)
  Form next state s'_{pvt} of the PVT agent using eq. (11)
  return (s, a, r, s')

according to (5). Second, we use the corner suggested by the PVT agent's actor and resimulate the circuit in that corner to obtain the reward r_2 according to (5). The reward for the PVT agent is then

$$\Delta = \begin{cases} 50 \times (r_1 - r_2); & r_1, r_2 > -0.5 \ \ \& \ \ |r_1 - r_2| < 0.1 \\ 10 \times (r_1 - r_2); & \text{otherwise} \end{cases}$$

$$\sigma(\Delta) = \frac{1}{1 + e^{-\Delta}} \rightarrow R_{\text{pvt}} = \begin{cases} \sigma, & \text{if } \Delta > 0 \\ \sigma - 1, & \text{if } \Delta < 0 \\ -1, & \text{if } \Delta = 0. \end{cases} \quad (10)$$

As can be seen in (10), we use the difference between the two rewards (Δ) as the input of a nonlinear sigmoid function. This sigmoid function assists in squashing the reward scale to a manageable range for all different scenarios of Δ. In our experiments, we found that in order for the PVT agent to function properly, three further reward shaping strategies are needed; 1) increasing the difference by a factor of 50 when both rewards are higher than -0.5 and the absolute difference is under 0.1 (indicating a relatively good design with a small difference in its performance across the two corners), and by a factor of 10 in all other cases (this calibration of the sigmoid slope is still needed to align it with the spectrum of reward differences that occur for different designs across the corners), 2) shifting the sigmoid output σ by -1 for a negative Δ to further punish the PVT agent (because negative Δ means that r_2 is greater than r_1, which indicates that the PVT corner

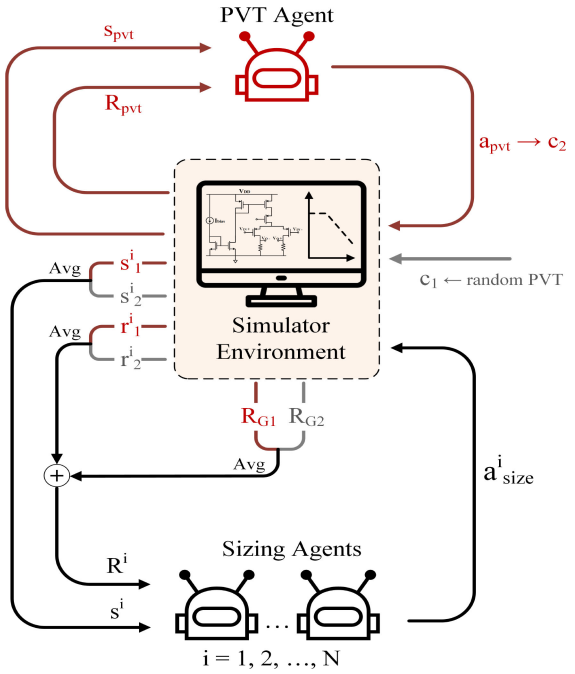


Fig. 2. Duel-Play RL sizing with N cooperative sizing agents and a competitive PVT agent. The state, reward and action of only one sizing agent are shown; each sub-block i has its corresponding R^i, s^i and a^i. Each R^i is the sum of average sub-block FoM in two corners and average total circuit FoM in two corners. Each s^i is the average of normalized performance metrics of the sub-block in two corners.

suggested by the PVT agent not only does not degrade the performance but in fact improves it), and 3) assigning a more negative value of -1 (instead of -0.5 if we follow $\sigma(0) - 1$) for cases when the performance is the same in the two corners ($\Delta = 0$) to indicate a mildly bad performance of the PVT agent and to prevent it from suggesting similar corners across multiple steps (encouraging more exploration). We really want the PVT agent to look for bad PVT corners.

PVT State Space: Each process-voltage-temperature corner comprises three discrete parameters.

1) *Process Parameters:* $\{SS, TT, FF\} \rightarrow \{-1, 0, 1\}$.
2) *Voltage:* $V_{DD} \pm 10\% \rightarrow \{-1, 0, 1\}$.
3) *Temperature:* $\{Low, Nominal, High\} \rightarrow \{-1, 0, 1\}$.

Therefore, each PVT corner is represented by a feature vector in $\{-1, 0, 1\}^3$, where each dimension corresponds to one of process-voltage-temperature. For instance, the corner [SS, 1.1V, 100 °C] maps to the feature vector $[-1, -1, 1]$. To construct the PVT state space at each step, we concatenate the feature vectors of the two PVT corners along with the sizing parameters at that step

$$s_{\text{pvt}} = [P_1, V_1, T_1, P_2, V_2, T_2, W_1, L_1, W_2, L_2, W_3, \ldots]. \quad (11)$$

This state representation encodes the most useful information for the PVT agent, based on which it can decide which new corner should be suggested as the action at each state.

PVT Action Space: This is a discrete space since we consider a finite number of corners as the action options. For instance, if we have three options for each of the process-voltage-temperature parameters, the action space has $3^3 = 27$

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Algorithm 3 Duel-Play Probabilistic-Model-Based Multiagent SAC for Sample-Efficient PVT-Robust Sizing

```
Initialize SAC for sizing agents and the PVT agent
Initialize replay buffers D_real and D_model,
Initialize probabilistic models P_ω
for each episode do
  if episode mod U = 0 then
    Train/Update global models P_ω on D_real
  end if
  for M model rollouts do
    Sample s_m uniformly at random from Dreal
    for k-steps do
      Take action a_m using the sizing agents
      Input (s_m, a_m) to P_ω; obtain s'_m and R_m
      Add transitions (s_m, a_m, r_m, s'_m) to Dmodel
      s_m ← s'_m {set rollout states to next rollout states}
    end for
  end for
  for each environment step do
    (s, a, r, s') ∼ Two_Corner_Simulation (s)
    Store transitions (s, a, r, s') in D_real
    if step > W warmup steps then
      for each sizing agent do
        Sample ϵB from D_real and (1−ϵ)B from D_model
        Update actor and critic networks of sizing agents
      end for
      Sample B transitions of the PVT agent from D_real
      Update actor and critic networks of the PVT agent
    end if
  end for
end for
Output: Optimized circuit sizing parameters
```

options in its discrete space. In order to handle this with SAC, we use SAC with discrete action settings as the PVT agent [29]; this was an extension of the original SAC agent which was developed mainly for continuous action spaces. The main difference with the original SAC is that the critic network outputs Q-values for all possible actions rather than for a single input action. Also, the actor network directly outputs discrete probabilities for all actions (instead of a mean and covariance of a continuous probability distribution) using a softmax function in its final layer.

2) Cooperative Sizing Agents: For the sizing agents in the duel-play scheme, their reward and state space should consider two corner simulations at each step.

Two-Corner Reward: Each agent i has reward values r_1^i and r_2^i from the simulations of its corresponding sub-block conducted in the two corners. Similarly, we define per-corner reward components for the entire circuit's performance concerning the overall specifications as R_{G1} and R_{G2}. First, to incorporate the results from both corner simulations into our reward calculations, we compute the average rewards for the two corners: $r^i = (r_1^i + r_2^i)/2$, $R_G = (R_{G1} + R_{G2})/2$. This approach, rather than taking the minimum, prevents agents from avoiding the neighborhood of promising designs due to poor performance in only one of the corners. Additionally, designs that succeed in one corner but fail in another are assigned a reward of 0 to discourage overfitting to nonrobust solutions. Second, for every sizing agent, we add the overall reward component to the agent's reward, treating this summed value as the final reward for that agent: $R^i = r^i + R_G$. These steps are illustrated in Fig. 2.

The above description is for the more general case of a complex circuit that has multiple sub-blocks. It should be noted that in this case, the r_1 and r_2 terms used for the PVT reward [in (10)] are: $r_1 = \Sigma r_1^i + R_{G1}$ $r_2 = \Sigma r_2^i + R_{G2}$. For simple single-block circuits, we only have the R_{G1} and R_{G2} terms, which we can simply consider as the r_1 and r_2 terms in both the sizing and PVT reward computations.

Two-Corner State Space: Similarly for the state space of each sizing agent (sub-block i) at each step, we take the element-wise average of the two state vectors (of the two corners); each of these is derived from the normalized specifications achieved after simulating at the corresponding corner according to (6): $s^i = (s_1^i + s_2^i)/2$.

D. Model-Based Policy Optimization

In MBPO, probabilistic neural networks are used as performance models to substitute for circuit simulations. These neural networks are designed to learn and provide the mean and the variance of a Gaussian distribution for each performance during training. As a result, they produce a distribution of possible outputs, rather than a single deterministic output, reflecting the uncertainty of the model prediction. Using an ensemble of these models (n_m models) for data generation can mitigate the epistemic uncertainty. Since PVT variations can be regarded as aleatoric uncertainty (due to their inherent irreducible nature which stems from variations in manufacturing processes and operating conditions), the above-mentioned probabilistic models can help to handle them. In our duel-play model-based SAC framework (Fig. 3 and Algorithm 3), we utilize these models for accelerating the policy optimization as well for ensuring the robustness against PVT variations. Each model is expected to receive the current state and the selected action (s, a) as input, and outputs the mean and variance of the predicted next state s'

$$p_\omega^i(s'|s,a) = \mathcal{N}\big(\mu_\omega^i(s,a), \Sigma_\omega^i(s,a)\big) \tag{12}$$

in which the mean μ_ω and the matrix of covariances Σ_ω are parameterized by ω as they are outputs of the model neural networks. Assuming a batch of N_b real transitions (s, a, r, s') from $\mathcal{D}_{\text{real}}$, if (s_n, a_n) is the state-action pair of each transition and $s_n\prime$ is its true next state (used as the true label for the output of the model), then the loss function used for training such model networks is

$$L_p(\omega) = \sum_{n=1}^{N_b} \Big[\big[\mu_\omega(s_n,a_n) - s_n'\big]^T \Sigma_\omega^{-1}(s_n,a_n) \big[\mu_\omega(s_n,a_n) - s_n'\big] + \log\det \Sigma_\omega(s_n,a_n) \Big]. \tag{13}$$

In the first term of this loss function, the mean errors are weighted inversely by their variances to penalize confident

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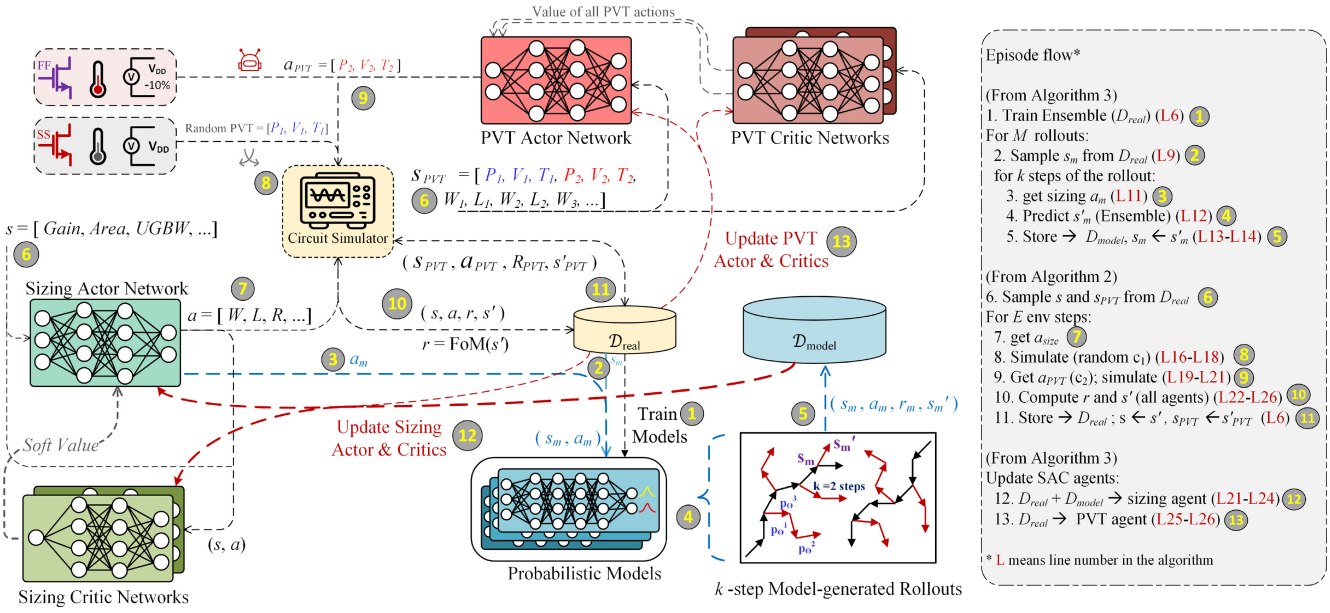


Fig. 3. Detailed diagram of the AnaCraft duel-play MBPO illustrated for the case of a single sizing agent and the adversarial PVT agent. The flow assumes that the initial warm-up steps are done and that there are initial data in D_{real}.

errors more. In the second term, the determinant of the variances is considered to discourage generally large variances, i.e., to promote predictions that are both accurate and confident.

In model-free SAC, after W warm-up steps (100 in our experiments), the Actor and Critic networks are updated using a batch of transitions (s, a, r, s') randomly sampled from the previous data stored in the replay buffer $\mathcal{D}_{\text{real}}$. In our duel-play MBPO, models are trained/updated once every U episodes (U is 10 in our experiments). Then, at the end of each episode, these models are used to roll out M short k-step trajectories starting from randomly selected previous transitions (M is 4000 and k is gradually increased from 1 to 3 steps during the learning process in our experiments). During each step of these k-steps, one of the models with lower L_p loss is selected at random, allowing for different transitions along a single rollout to be sampled from different dynamic models. These model-generated transitions are stored in another replay buffer $\mathcal{D}_{\text{model}}$. Then, at each policy optimization step, a combination of real and generated transitions from the two replay buffers is selected as the mini-batch ($\mathcal{B}$) of data for updating the Actor and Critic networks. In other words, for a better handling of the bias of the models, at each policy optimization step, MBPO selects a small portion $\epsilon\mathcal{B}$ of the batch from the real data ($\mathcal{D}_{\text{real}}$), and the remaining portion of the batch, $(1-\epsilon)\mathcal{B}$, is chosen from the model-generated data $\mathcal{D}_{\text{model}}$ ($\mathcal{B}$ and ϵ are 100 and 0.2 in our experiments). This approach of using generated trajectories significantly assists the agent in observing unexplored spaces faster and therefore to speed up the policy optimization. Algorithm 3 explains the different steps of the duel-play MBPO: U, M, and k are hyperparameters to be tuned in this algorithm. However, the usage of SAC helped us in expediting the hyperparameter tuning process. We trained and updated an ensemble of seven neural networks (each having four hidden layers of 200 neurons) and picked the five models with the lowest L_p loss for the rollout generation.

In the case of complex circuits that have multiple sub-blocks, we use one ensemble of global models. They predict the next state of each sub-block, as well as of the total circuit, at their output. Empirical results show that having global models is sufficient and helps in the scalability of our algorithm toward more complex circuits. It should be noted that we do not use a model for the adversarial PVT agent; this is not needed since the model-free SAC agent is sufficient for finding challenging corners. It is mainly the job of the probabilistic models of the sizing agents to accelerate the handling of the PVT variations and to find sizes that offer PVT-robust performances.

Although circuit sizing is not inherently episodic, we employ fixed-length episodes ($E = 10$ environment steps in our implementation) for improved learning stability. An "environment step" involves action selection by all agents, the necessary simulations (typically two in duel-play) and storing the transition in $\mathcal{D}_{\text{real}}$. Upon completing E steps, the episode terminates and restarts from a state sampled from the replay buffer. As indicated in Algorithms 1–3, SAC agent networks are updated at each environment step using sampled data.

IV. Experimental Results

To evaluate the performance of our AnaCraft algorithm, we have performed two sets of experiments. In the first set, we have sized two operational amplifiers in a predictive 45-nm CMOS technology. We use predictive technology to improve reproducibility, considering its free accessibility for everyone. We show the superiority of SAC over TD3, and of duel-play SAC over model-free SAC and TD3 for PVT-robust sizing. We provide a progressive ablation study of our

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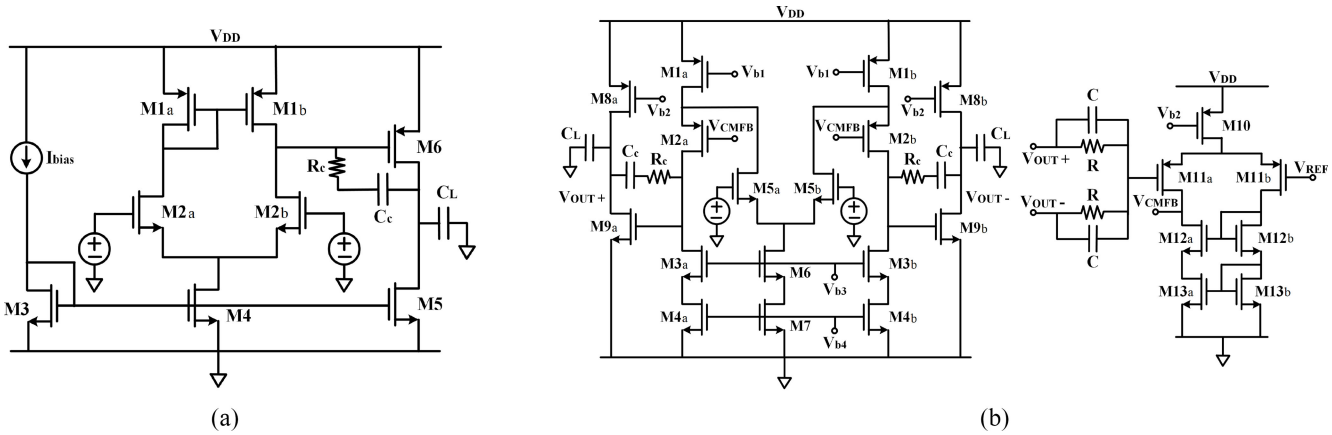


Fig. 4. Circuit Schematics of the operational amplifiers used in the experiments. (a) Two-stage opamp. (b) Differential folded-cascode opamp.

algorithm to show the importance and effectiveness of each of its components in achieving optimal and robust solutions. Then, in the second set of experiments, we have sized a complex data receiver circuit in a predictive 7-nm FinFET technology, and we show that our duel-play multiagent model-based SAC outperforms both a human reference design and the previous state-of-the-art methods for the sizing optimization of complex circuits with PVT-robustness. We then present a final discussion on why our method is superior to other existing methods.

A. Sizing Opamps in 45-nm CMOS Predictive PDK

We start our experiments by sizing two distinct operational amplifiers in a 45-nm BSIM predictive technology [30]. For these experiments, we use NgSPICE [31] as the circuit simulator. For the area calculation, we consider an estimate of the active area for each set of actions. We run the algorithms on a server with Intel Xeon Silver 4110 CPU cores and the Nvidia GeForce RTX 2080ti GPU. The two amplifier circuits are: 1) a basic two-stage opamp and 2) a differential folded-cascode opamp with common-mode feedback (see Fig. 4).

For these circuits, we consider five hard specification constraints: 1) Gain; 2) Phase Margin; 3) Unity Gain Bandwidth; 4) Slew Rate; and 5) Output Noise; and we aim for the minimization of the total current (or power consumption) and the area. We search for the optimum transistor sizing parameters (W, L, m), resistor values (R) and capacitor values (C) in both circuits and, for the folded-cascode opamp, also for the biasing voltages. The ranges of the different design variables are: $W = [0.25, 5]$ μm, $L = [45, 225]$ nm, $m = [1, 25]$, $C = [0.1, 10]$ pF, Miller resistor $R_c = [100, 10k]$ Ω, CMFB resistor $R = [1k, 1M]$ Ω, and the biasing voltages $V_i = [0, 1.2]$ V. Note that V_{DD} is 1.2 V. The reference current $ibias$ is 30 μA, and the capacitive loads C_L are 10 pF. Matching constraints (e.g., in differential pairs and current mirror loads) are enforced by assigning identical parameters (W, L, m) to matched transistors. Nonetheless, the basic two-stage opamp has 20 optimization parameters, while the folded-cascode opamp has 48 optimization parameters. We performed 100 random simulations, which served as the warm-up population for all algorithms. The t^* normalization values (4) were obtained by averaging the current/area of the top three designs from these simulations. These are helper values that are used instead of using $t^* = 0$, in order to avoid the corresponding normalized values of power and area to become always 1 when using (4).

We consider 27 PVT corners when sizing these opamps: they consist of {TT, FF, SS} $\times$ {1.2 V, 1.1 V, 1.0 V} $\times$ {0 °C, 27 °C, 100 °C}. In our experiments, initially, for the first 10 instances where we find two-corner solutions, we sequentially evaluate other corners following the probabilities given by the PVT agent's Actor network output. This approach avoids simulating all corners for solutions that are not fully PVT-robust. Since we assign a reward of 0 to these designs, the PVT agent gradually improves in identifying full PVT-robust solutions rather than just two-corner solutions, reducing the number of corners that might fail the candidate solutions. Therefore, after ten samples of finding two-corner solutions, we proceed by simulating and evaluating all remaining corners in parallel for each new candidate solution. This consideration helps to reduce the runtime. Nonetheless, all our plots include all simulations that are performed throughout the entire algorithm execution.

1) Sizing Results: Fig. 5 shows a comparison of the different algorithms in terms of the maximum FoM reached over simulation steps. In this figure, we present a comparison of our duel-play multiagent MBPO approach (AnaCraft) against several other methods. These include ablation studies designed to highlight the significance of each component of our algorithm. Furthermore, we have implemented and employed RobustAnalog [13], an existing RL-based PVT-robust sizing method, as a baseline to benchmark our approach. We first run the state-of-the-art MATD3 with two random corners per step (i.e., RandomPVT MATD3) to see the impact of considering two distinct corners at each step. Then, we equip MATD3 with the duel-play scheme to show the improvement that we can get from the competition of the sizing and PVT agents (Duel-Play MATD3). Next, we change the agent from TD3 to SAC (Duel-Play MASAC). Finally, we add the probabilistic models and rollouts to construct the full duel-play AnaCraft.

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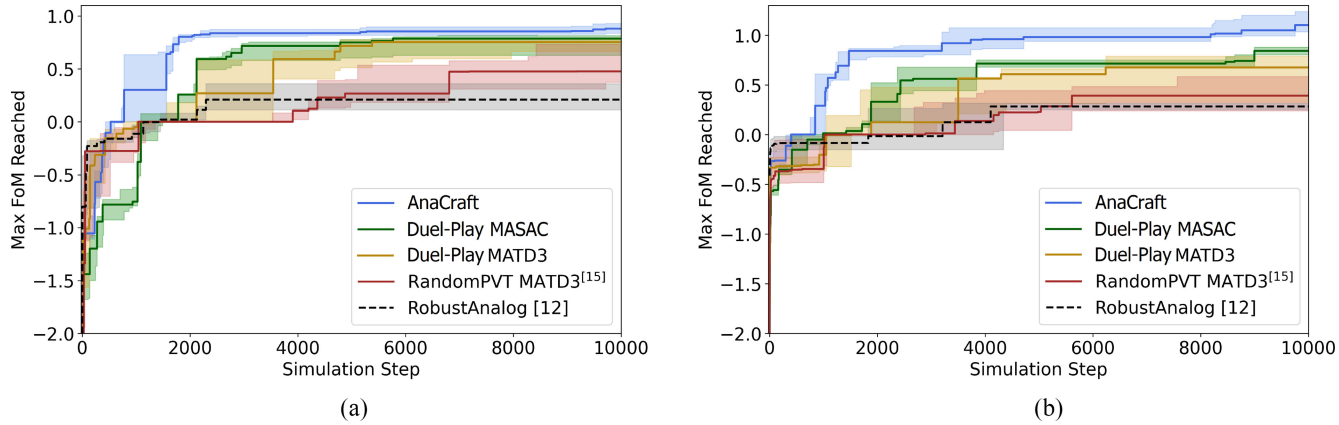


Fig. 5. Comparison of the maximum FoM reached per step for the two demonstrator opamp circuits obtained by different sizing algorithms. (a) Two-stage opamp. (b) Differential folded-cascode opamp.

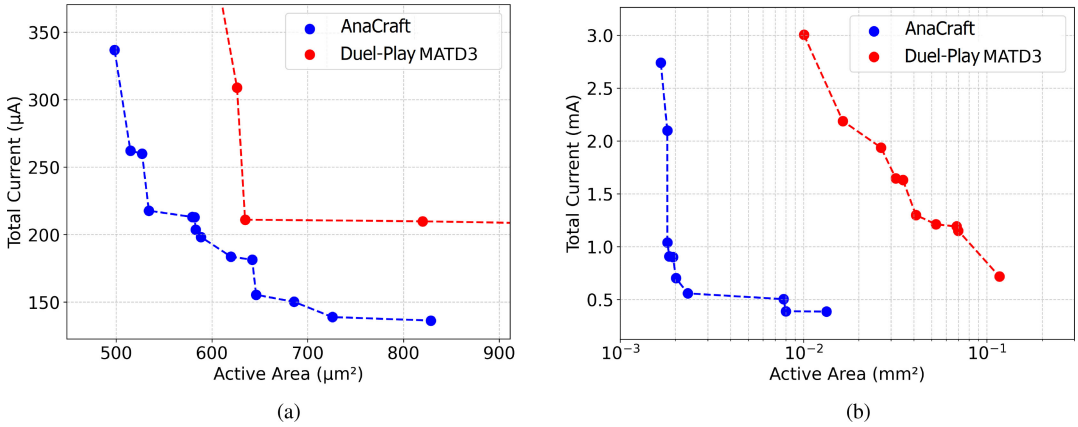


Fig. 6. Comparison of the (Pareto-)optimal fronts for the two demonstrator opamp circuits obtained by AnaCraft and Duel-Play MATD3. (a) Two-stage opamp. (b) Differential folded-cascode opamp.

We run each algorithm five times and show the median performance as a bold line while using a shadow color to indicate the variability of the performances across different runs.

From these plots, it can be seen that merely simulating the circuit in two random corners at each step and considering the average performance already can lead to robust solutions. We also observe that the performance of RandomPVT MATD3 is comparable to that of RobustAnalog for the differential folded-cascode opamp. It can also be seen that the two TD3-based algorithms show more variability across different runs (wider shadows), while the SAC-based algorithms show more stability in their performance. This is because SAC considers entropy maximization in its soft Q-function (as discussed in Section III). Most importantly, the duel-play AnaCraft is considerably faster than all other baseline algorithms in both finding a good solution fast and in further optimizing the power and area. In particular, AnaCraft is on average $\sim 3\times$ faster than RobustAnalog in terms of the number of simulations needed until finding the first full PVT-robust solution.

Furthermore, in Fig. 6, we show the comparison of the (Pareto-)optimal solutions achieved by our AnaCraft method and the duel-play MATD3. It is clear from these plots that the probabilistic models and the SAC agent are highly effective in finding diverse power-area optimal solutions to derive these optimal fronts.

TABLE I
PERFORMANCE COMPARISON OF THE OPAMP SOLUTIONS GENERATED WITH OUR ANACRAFT AND A BASE DUEL-PLAY MATD3

Performance	two-stage opamp			folded-cascode opamp		
	target	base	ours	Target	base	ours
Gain	200	217	292	350	461	380
Phase Margin (°)	60	82	63	60	60.8	78
Unity GBW (MHz)	100	111	115	200	445	289
Slew Rate ($V/\mu sec$)	15	18.4	19.3	5	45.3	16.3
Out. Noise (mV_{rms})	30	5.2	5.5	50	44.5	36.3
Total Current (mA)	0.2*	0.28	0.23	4*	3.27	1.85
Area ($\times 10^{-3} mm^2$)	2*	0.89	0.95	30*	21	35
FoM	-	0.51	**0.57**	-	0.55	**0.58**
# Simulations	-	3908	**1478**	-	3506	**1082**
Runtime ($hours$)	-	1.57	**1.16**	-	2.5	**1.218**

* Soft target values (t^* in eq. (4)) used only for normalization

Finally, in Table I, we show the comparison of the number of simulations, the runtime, and the performances reached for the first solution that has a FoM above 0.5 for each opamp. Note that these points are considered for comparison purposes. It is evident from the plots in Fig. 5 that changing this reference FoM point does not change the performance comparison

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TABLE II
COMPARISON OF THE FRACTION OF THE GENERATED SOLUTIONS THAT PASS ALL PVT CORNERS

opamp	RandomPVT MATD3	Duel-Play MASAC
two-stage	8%	73%
folded-cascode	3%	68%

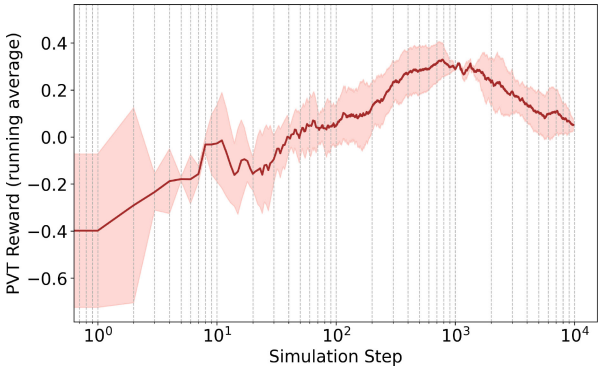


Fig. 7. Running average of the PVT agent reward over the steps of the Duel-Play MASAC algorithm.

numbers significantly. It should be noted that RobustAnalog achieves the first PVT-robust solution in 1.61 (2.35) h and 3919 (3314) simulations for the two-stage (differential folded-cascode) opamp, on average. Since the duel-play MATD3 achieves higher FoMs with almost similar speed, we have selected it as a stronger baseline for this table. It can be seen that our AnaCraft algorithm is 1.35× and 2× faster than the base duel-play MATD3 algorithm (and by extension, faster than RobustAnalog) in terms of runtime for the two-stage opamp and the folded-cascode opamp, respectively. Although the increased computational demands associated with model training lead to smaller runtime enhancements compared to the reductions in simulation count, this overhead becomes relatively less significant for larger circuits, since the simulation durations are longer.

2) PVT Agent Analyses: To further show the effectiveness of the adversarial PVT agent, we first compare the ratio of solutions that pass all PVT corners after having passed the two corners. Then we show an example of the learning curve for the adversarial PVT agent throughout the steps of the algorithm.

As can be seen in Table II, when we only simulate the circuit in two random corners and collect the solutions after 10k simulations, on average less than 10% of them are robust across all remaining PVT corners. However, when we use the PVT agent to choose the adversarial corner at each step, more than 60% of the solutions found pass all remaining corners. This result shows that the PVT agent is indeed significantly better in finding bad corners and therefore in helping to find full PVT-robust solutions.

Moreover, although we want both types of agents to be trained well, it is needless to say that the end goal for the sizing agents is to defeat the PVT agent. In other words, through time we want our sizing agents to find sizing parameters that are solutions to the analog circuit optimization no matter which two corners are simulated. As can be seen from Fig. 7,

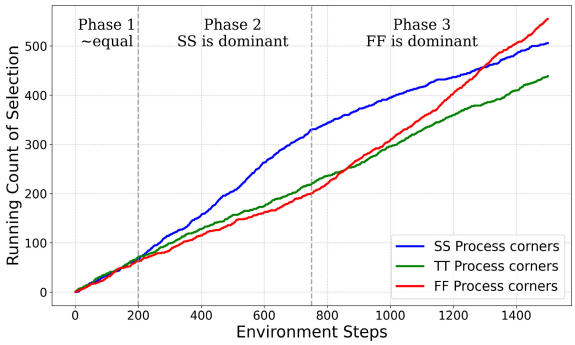


Fig. 8. Example of the running count of different process corners suggested by the PVT agent through steps of the algorithm.

which shows the reward of the PVT agent when sizing the two-stage opamp, during the steps of the Duel-Play MASAC RL algorithm, the running average reward of the PVT agent increases in the beginning; which means that the agent is learning to find difficult corners for each situation. However, after ∼1000 simulation steps, the average reward goes down again, which indicates that the sizing agent(s) have started to find more robust (PVT-agnostic) solutions.

It is crucial to interpret Table II correctly: the ∼70% success rate for Duel-Play MASAC demonstrates the effectiveness of the two-corner check as a strong predictor of full PVT robustness, a significant improvement over the 3%–8% success rate using two random corners. This high correlation indicates that the PVT agent successfully identifies relevant challenging corners for the *current* design state, rather than trying to find a single, universal worst-case corner.

The PVT agent's objective is adaptive; it learns to propose corners that maximize its reward (10) based on the performance difference relative to the random corner, considering the specific design parameters being evaluated. This encourages exploration of diverse failure modes. For instance, analysis of the agent's behavior during the two-stage opamp optimization shows a dynamic strategy: initially, it frequently challenges the sizing agent with SS corners, but as the sizing agent adapts and improves the robustness against SS variations (making SS corners less rewarding for the PVT agent), it shifts focus to probe other potentially critical corners, such as FF (as shown in Fig. 8). This adaptive stress-testing forces the sizing agent toward global robustness.

B. Sizing a Data Receiver Circuit in the ASAP 7-nm FinFET PDK

In order to show the ability of our algorithm for truly complex circuits in advanced technologies, we consider sizing a data receiver circuit in the ASAP 7nm FinFET technology [33]. This circuit is designed to be used inside a physical interface (PHY) that connects a System on Chip (SoC) to a dynamic random access memory (DRAM) with LPDDR4X standard. The receiver circuit amplifies and recovers the attenuated data signal transmitted by the DRAM. The circuit comprises three sub-blocks and must function at a maximum frequency of 2.133 GHz. The other hard constraints are summarized in Table III. The full receiver topology starts with a

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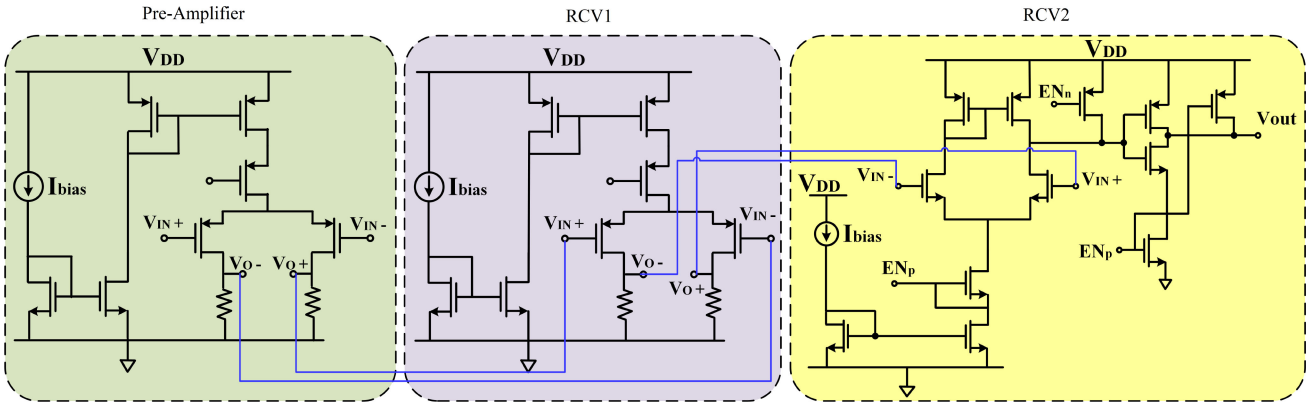


Fig. 9. Schematics of the entire data receiver circuit and its sub-blocks: Preamplifier, RCV1, and RCV2.

preamplifier block, which converts the single-ended signal into a differential signal. The preamplifier should have a gain above 1 and a unity gain-bandwidth above ∼20 MHz to not attenuate the signal even more. The second block is the receiver RCV1 which is the first amplification stage, and the third block is the receiver RCV2 (with a NAND gate at its output) that must amplify the signal to full swing to capture it by internal registers; its gain is to be maximized while maintaining a unity gain-bandwidth above ∼10 MHz. The schematic of the circuit is shown in Fig. 9. The transistors in this circuit are FinFET transistors and (as mentioned in Section III) their parameters are the length $L = \{7n, 16n, 20n\}$, the number of fins $N^{fin} = [1, 100]$, and the number of fingers $N_f = [1, 10]$. The *Ibias* is 50 μA and given by an internal bandgap circuit. The circuit has 41 optimization parameters in total. Similar to the previous experiments, we consider 27 corners consisting of {TT, FF, SS} × {0.8 V, 0.88 V, 0.72 V} × {−40 °C, 25 °C, 125 °C}. We use Cadence® SPECTRE [32] as the simulator for this circuit.

There is one other consideration which makes this circuit more challenging: it must function in two input scenarios.

1) *Unterminated Case:* Maximum swing of $V_{\text{in}} = 0.6\ V$ with a V_{ref} of 0.3 V.
2) *Terminated Case:* Maximum swing of $V_{\text{in}} = 0.3\ V$ with a V_{ref} of 0.15 V.

To handle this, we randomly pick one scenario at each step and only simulate the second scenario if the first one gives a solution; we average the performance across the two input scenarios in that case.

Similarly to the previous experiments, in Fig. 10(a), we show the comparison of the different algorithms in terms of the maximum FoM reached per simulation step. It should be noted that for these plots we show the average FoM (reward) only when all the sub-blocks and the total circuit pass their hard constrained specifications in both corners and subsequently in all remaining corners. In cases where the specifications (per-block and general) are satisfied only in some corners but not in all corners, we show a FoM of zero. Therefore, in these plots (as well as in the plots of Fig. 5), the maximum FoM curve stays on zero for a number of steps until it finally finds the full PVT-robust solution and passes beyond zero.

TABLE III
PER-BLOCK DESIGN CONSTRAINTS OF THE DATA RECEIVER AND GENERAL PERFORMANCE RESULTS

Spec.	Pre-Amp.	RCV1	RCV2
Unity GBW (*GHz*)	> 20	> 20	> 20
Gain	> 1	> 1	Maximize

Performance	Target	Duel-Play MATD3	AnaCraft
Unity GBW (*GHz*)	> 10	110	137
Slew Rate (*V/nsec*)	> 10	27	29
Total Gain	1.0*	219	473
Total Current (*mA*)	0.7*	0.7	1.1
Area (μm^2)	15*	16.8	8.1
FoM	-	0.51	**0.54**
#Simulations	-	3416	**1233**
Runtime (hours)	-	3.42	**1.76**

As evidenced by the plots, our duel-play AnaCraft algorithm can reach the first solution within ∼1000 simulations, while the other algorithms need more than 2000–3000 simulations. AnaCraft also finds higher values of the FoM than the other algorithms can reach. In particular, it finds solutions with a higher FoM value than that of the reference solution (designed by a human designer) with less than 2000 simulations. Also, it can be seen again that the SAC-based algorithms show less fluctuations and variance across different runs. For this circuit, RandomPVT MATD3 cannot reach a solution within 5000 simulations.

There are three optimization targets for this circuit: 1) power; 2) area; and 3) gain. The gain maximization is achieved by maximizing the gain of the last stage (RCV2). When adding the reward term corresponding to the gain [following (5)], we consider a weight of 0.0005 for it and limit its maximum reward contribution to 0.3. Considering this, we have three (Pareto-)optimal plot comparisons. Fig. 10(b), (c), and (d) show the power-area, the gain-power and the gain-area Pareto tradeoffs, respectively. It can be seen that AnaCraft can find solutions that are better in all of the optimization targets compared to the Duel-Play MATD3, and by extension better than the algorithms used in the state-of-the-art tools. In particular, compared to the human reference design, we found solutions that have a similar range of gain (or even higher) with significantly less total current and much smaller area.

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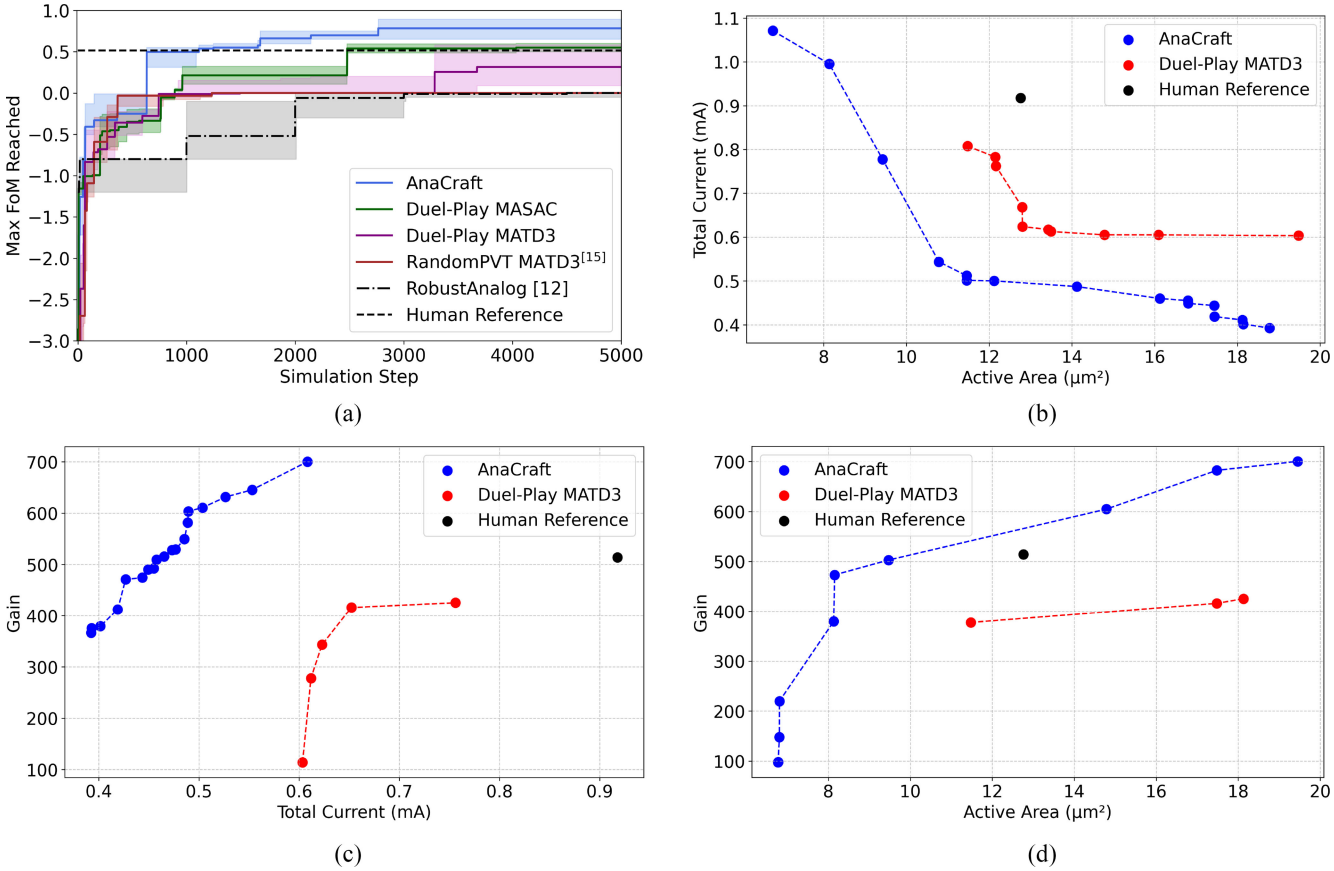


Fig. 10. Comparisons of different sizing algorithms for optimizing the data receiver circuit in the ASAP 7-nm FinFET technology. (a) Maximum reached FoM per step. (b) Current-area (Pareto-)optimal fronts. (c) Gain-current (Pareto-)optimal fronts. (d) Gain-area (Pareto-)optimal fronts.

Finally, Table III shows a comparison of AnaCraft and the Duel-Play MATD3 in terms of number of simulations, runtime, and the achieved performances at the first point where the algorithm finds a solution with a FoM above 0.5. As seen, the AnaCraft algorithm is ∼3× and ∼2× faster than Duel-Play MATD3 in terms of simulation count and runtime, respectively.

TABLE IV
RUNTIME BREAKDOWN OF ANACRAFT (IN MINUTES) AFTER 1478, 1082, AND 1233 SIMULATIONS FOR THE TWO-STAGE OPAMP, THE FOLDED-CASCODE OPAMP, AND THE DATA RECEIVER, RESPECTIVELY

Algorithm Part	Two-stage	Folded-cascode	Data Receiver
Real circuit simulations	19 (27.1%)	34 (46.5%)	50 (47.1%)
Model Ensemble Train	33 (47.1%)	26 (35.6%)	31 (29.2%)
SAC Agents' Updates	16 (22.8%)	11 (15%)	21 (19.8%)
Rollout Generations	0.9 (1.3%)	0.6 (0.8%)	2.2 (2%)
Total runtime	70	73	106

C. Runtime Cost Analysis

To provide insight into the computational cost associated with our approach, Table IV details the runtime breakdown for AnaCraft across the three tested circuits. The primary components include the real circuit simulations, the probabilistic model ensemble training/updates, the actor–critic network updates for all agents, and the model-based rollout generation. While incorporating MBPO introduces overhead compared to purely model-free methods, this cost enables substantial reductions in the most important part: the real circuit simulations. As observed by comparing the relative costs, the overhead of the MBPO model training and usage becomes less compared to the simulation time when the circuit complexity and simulation time per step increase (e.g., compare the two-stage opamp versus the data receiver), which is consistent with the findings reported in the previous section. The cost of generating rollouts remains minimal ($< 2\%$).

D. Comparison to the State-of-the-Art

As seen in the previous sections, assigning a competitive agent to the task of choosing challenging PVT corners can help the sizing agents to arrive at PVT-robust solutions. This is due to the fact that the PVT agent learns more about the variations of the performances across the PVT corners. Therefore, through time, it gradually learns which difficult corners to choose in different sizing regions. On the other hand, the sizing agents, which are equipped with probabilistic performance models, gradually guide the search to sizing regions where the performances are satisfied regardless of which PVT corner is simulated. Therefore, our method can make the whole task more efficient and effective by adding only one extra simulation per step.

Prior works require more than two simulations per step due to less inference and learning of the behavior of the

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circuit across different corners. RobustAnalog [13] introduced K-means clustering and multitask RL for finding PVT-robust solutions (also adopted by [19]). It classifies the different PVT corners based on circuit performance using K-means clustering and selects a representative subset of challenging corners for simulation at each step. While this approach does handle PVT variations, it suffers from sample inefficiency for two main reasons: 1) the clustering is based on a single set of sizing parameters at each iteration, which leads to limited PVT inference capabilities and 2) it requires 4–5 simulations per step, depending on the number of clusters.

On the other hand, approaches such as RoSE [21] and PVTSizing [19] utilize BO to enhance the initial data collection or presearch steps. BO, however, becomes computationally expensive for complex circuits with many parameters and is challenging to integrate into MARL frameworks. This is in contrast to our work, which has more potential for scalability to large circuits by adapting to the MARL scheme for handling the different requirements of the sizing task.

To provide a direct quantitative comparison, we benchmark AnaCraft against PVTSizing [19] for our three circuits. Table V compares the runtime and the total number of simulations required by each algorithm to find the first fully PVT-robust solution (on the same server as previous experiments). We compare based on finding the first robust solution as PVTSizing is primarily designed for this goal (similar to RobustAnalog), after which it stops. The results show that AnaCraft consistently achieves this goal faster and with fewer simulations. This advantage is particularly evident in the complex data receiver case, where AnaCraft requires $\sim$2.7$\times$ less runtime and $\sim$4.5$\times$ fewer simulations. We observed that PVTSizing's initial phase, which uses BO (TuRBO) to find nominal solutions before PVT considerations, can become a bottleneck for complex circuits due to BO's scaling limitations.

The efficiency gains of AnaCraft are attributed to its handling of PVT corners and its model usage. PVTSizing's workflow involves steps that can increase the simulation cost, such as simulating multiple nominal-corner solutions across the full PVT set and employing critic-assisted pruning based on predicting the FoM across all corners—a complex task for the critic network. In contrast, AnaCraft's duel-play mechanism focuses simulations on only two corners per step, guided by a dedicated PVT agent learning which are the challenging corners, rather than requiring predictions across many corners simultaneously for pruning.[1]

Some other machine learning approaches focus on building accurate surrogate models of the circuit performance, such as the connectivity-inspired neural networks (CCI-NN) [34], which require structural circuit knowledge (e.g., signal sequential path, open-loop representation for feedback) for model construction. In contrast, AnaCraft directly targets the sizing optimization challenge using a novel RL framework without requiring any knowledge about the specific circuits being sized.

[1]The PVTSizing results are based on our implementation following [19]. We believe that the core algorithmic differences discussed, rather than the specific implementation details, primarily account for the observed performance variations.

TABLE V
COMPARISON OF RUNTIME AND SIMULATION COUNT NEEDED FOR FINDING THE FIRST FULL PVT-ROBUST SOLUTION

Circuit	PVTSizing		AnaCraft	
	Runtime (hours)	Total # Simulations	Runtime (hours)	Total # Simulations
Two-stage opamp	1.76	1865	1.06	1053
Folded-cascode	1.45	1428	1.14	920
Data Receiver	3.59*	3681*	1.32	826

* success rate of PVTSizing for the Data Receiver is 2 out of 5 runs

It can be concluded that AnaCraft is a more promising approach for PVT-Robust sizing optimization of complex analog circuits.

V. CONCLUSION

This article has presented the AnaCraft duel-play model-based multiagent approach as a faster and more robust reinforcement-learning-based automated analog circuit sizing algorithm. The cooperative sizing agents for the different circuit sub-blocks compete against an adversarial PVT agent to obtain PVT-robust circuit sizing solutions. In addition, an ensemble of probabilistic models is used to generate rollouts of data in order to find optimal solutions faster. The scalability of the algorithm to more complex circuits consisting of multiple sub-blocks and to advanced FinFET technologies has been demonstrated experimentally. It has been shown that the proposed algorithm obtains better and PVT-robust solutions faster than other state-of-the-art algorithms. This paves the way toward more efficient and more robust design automation of analog integrated circuits.

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REFERENCES

[1] M. Ahmadzadeh and G. Gielen, "Using probabilistic model rollouts to boost the sample efficiency of reinforcement learning for automated analog circuit sizing," in *Proc. 61st ACM/IEEE Design Autom. Conf. (DAC)*, 2024, pp. 1–6.

[2] M. F. M. Barros, J. M. C. Guilherme, and N. C. G. Horta, *Analog Circuits and Systems Optimization Based on Evolutionary Computation Techniques*, vol. 9. Heidelberg, Germany: Springer, 2010.

[3] G. Wolfe and R. Vemuri, "Extraction and use of neural network models in automated synthesis of operational amplifiers," *IEEE Trans. Comput.-Aided Design Integr. Circuits Syst.*, vol. 22, no. 2, pp. 198–212, Feb. 2003.

[4] T. Pessoa, N. Lourenço, R. Martins, R. Póvoa, and N. Horta, "Enhanced analog and RF IC sizing methodology using PCA and NSGA-II optimization kernel," in *Proc. Design, Autom. Test Europe Conf. Exhibition (DATE)*, 2018, pp. 660–665.

[5] W. Lyu, F. Yang, C. Yan, D. Zhou, and X. Zeng, "Batch Bayesian optimization via multi-objective acquisition ensemble for automated analog circuit design," in *Proc. 35th Int. Conf. Mach. Learn.*, 2018, pp. 3306–3314.

[6] B. Liu, D. Zhao, P. Reynaert, and G. G. E. Gielen, "GASPAD: A general and efficient mm-Wave integrated circuit synthesis method based on surrogate model assisted evolutionary algorithm," *IEEE Trans. Comput.-Aided Design Integr. Circuits Syst.*, vol. 33, no. 2, pp. 169–182, Feb. 2014.

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[7] K. Settaluri, A. Haj-Ali, Q. Huang, K. Hakhamaneshi, and B. Nikolic, "AutoCkt: Deep reinforcement learning of analog circuit designs," in *Proc. Design, Autom. Test Europe Conf. Exhibition (DATE)*, 2020, pp. 490–495.
[8] B. Shahriari, K. Swersky, Z. Wang, R. P. Adams, and N. De Freitas, "Taking the human out of the loop: A review of Bayesian optimization," *Proc. IEEE*, vol. 104, no. 1, pp. 148–175, Jan. 2016.
[9] P. I. Frazier, "A tutorial on Bayesian optimization," 2018, *arXiv:1807.02811*.
[10] H. Wang, J. Yang, H.-S. Lee, and S. Han, "Learning to design circuits," 2018, *arXiv:1812.02734*.
[11] H. Wang et al., "GCN-RL circuit designer: Transferable transistor sizing with graph neural networks and reinforcement learning," in *Proc. 57th ACM/IEEE Design Autom. Conf. (DAC)*, 2020, pp. 1–6.
[12] N. S. K. Somayaji, H. Hu, and P. Li, "Prioritized reinforcement learning for analog circuit optimization with design knowledge," in *Proc. 58th ACM/IEEE Design Autom. Conf. (DAC)*, 2021, pp. 1231–1236.
[13] W. Shi et al., "Robustanalog: Fast variation-aware analog circuit design via multi-task RL," in *Proc. ACM/IEEE Workshop Mach. Learn. CAD*, 2022, pp. 35–41.
[14] M. Choi, Y. Choi, K. Lee, and S. Kang, "Reinforcement learning-based analog circuit optimizer using gm/ID for sizing," in *Proc. 60th ACM/IEEE Design Autom. Conf. (DAC)*, 2023, pp. 1–6.
[15] K.-E. Yang et al., "Trust-region method with deep reinforcement learning in analog design space exploration," in *Proc. 58th ACM/IEEE Design Autom. Conf. (DAC)*, 2021, pp. 1225–1230.
[16] J. Zhang, J. Bao, Z. Huang, X. Zeng, and Y. Lu, "Automated design of complex analog circuits with multiagent based reinforcement learning," in *Proc. 60th ACM/IEEE Design Autom. Conf. (DAC)*, 2023, pp. 1–6.
[17] T. Haarnoja et al., "Soft actor-critic algorithms and applications," 2018, *arXiv:1812.05905*.
[18] J. Lan et al., "Effective gain analysis and statistic based calibration for ring amplifier with robustness to PVT variation," *IEEE Trans. Circuits Syst. II, Exp. Briefs*, vol. 69, no. 2, pp. 304–308, Feb. 2021.
[19] Z. Kong, X. Tang, W. Shi, Y. Du, Y. Lin, and Y. Wang, "PVTSizing: A TuRBO-RL-based batch-sampling optimization framework for PVT-robust analog circuit synthesis," in *Proc. 61st ACM/IEEE Design Autom. Conf. (DAC)*, 2024, pp. 1–6.
[20] A. L. Zimpeck, C. Meinhardt, G. Posser, and R. Reis, "FinFET cells with different transistor sizing techniques against PVT variations," in *Proc. IEEE Int. Symp. Circuits Syst. (ISCAS)*, 2016, pp. 45–48.
[21] J. Gao, W. Cao, and X. Zhang, "RoSE: Robust analog circuit parameter optimization with sampling-efficient reinforcement learning," in *Proc. 60th ACM/IEEE Design Autom. Conf. (DAC)*, 2023, pp. 1–6.
[22] A. F. Budak, P. Bhansali, B. Liu, N. Sun, D. Z. Pan, and C. V. Kashyap, "DNN-OPT: An RL inspired optimization for analog circuit sizing using deep neural networks," in *Proc. 58th ACM/IEEE Design Autom. Conf. (DAC)*, 2021, pp. 1219–1224.
[23] Y. Choi, S. Park, M. Choi, K. Lee, and S. Kang, "MA-Opt: Reinforcement learning-based analog circuit optimization using multi-actors," *IEEE Trans. Circuits Syst. I, Reg. Papers*, vol. 71, no. 5, pp. 2045–2056, May 2024.
[24] D. Silver et al., "Mastering the game of Go with deep neural networks and tree search," *Nature*, vol. 529, no. 7587, pp. 484–489, Jan. 2016.
[25] D. Silver et al., "A general reinforcement learning algorithm that masters chess, shogi, and Go through duel-play," *Science*, vol. 362, no. 6419, pp. 1140–1144, Dec. 2018.
[26] O. Vinyals et al., "Grandmaster level in StarCraft II using multi-agent reinforcement learning," *Nature*, vol. 575, no. 7782, pp. 350–354, 2019.
[27] K. Chua, R. Calandra, R. McAllister, and S. Levine, "Deep reinforcement learning in a handful of trials using probabilistic dynamics models," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 31, 2018, pp. 1–12.
[28] M. Janner, J. Fu, M. Zhang, and S. Levine, "When to trust your model: Model-based policy optimization," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 32, 2019, pp. 1–12.
[29] P. Christodoulou, "Soft actor-critic for discrete action settings," 2019, *arXiv:1910.07207*.
[30] W. Zhao and Y. Cao, "New generation of predictive technology model for sub-45nm early design exploration," *IEEE Trans. Electron Devices*, vol. 53, no. 11, pp. 2816–2823, Nov. 2006.
[31] H. Vogt, G. Atkinson, P. Nenzi, and D. Warning, "NgSPICE User's Manual Version 43 (NgSPICE Release Version)," 2024. [Online]. Available: https://ngspice.sourceforge.io/docs/ngspice-43-manual.pdf
[32] *Spectre Circuit Simulator, Version 21.1.0.132*, Cadence Design Syst., Inc., San Jose, CA, USA, 2021.
[33] L. T. Clark et al., "ASAP7: A 7-nm finFET predictive process design kit," *Microelectron. J.*, vol. 53, pp. 105–115, Jul. 2016.
[34] M. Hassanpourghadi, S. Su, R. A. Rasul, J. Liu, Q. Zhang, and M. S.-W. Chen, "Circuit connectivity inspired neural network for analog mixed-signal functional modeling," in *Proc. 58th ACM/IEEE Design Autom. Conf. (DAC)*, 2021, pp. 505–510.



Mohsen Ahmadzadeh received the B.Sc. degree in electrical engineering from the K. N. Toosi University of Technology, Tehran, Iran, in 2018, and the M.Sc. degree in electronics from the University of Tehran, Tehran, in 2021. He is currently pursuing the Ph.D. degree with the MICAS Research Group, Department of Electrical Engineering, KU Leuven, Leuven, Belgium.

His research interests are machine learning and computer-aided IC design.



Jan Lappas (Member, IEEE) received the B.S. and M.S. degrees in electrical engineering from the Technical University of Kaiserslautern, Kaiserslautern, Germany, in 2015 and 2017, respectively, where he is currently pursuing the Ph.D. degree with the Microelectronic Systems Design Research Group.

His research interests are mixed-signal chip design, especially with the focus on cross-layer optimization of memory and communication systems.



Norbert Wehn (Senior Member, IEEE) received the Ph.D. degree (summa cum laude) in VLSI design and CAD for microelectronics from the Technical University of Darmstadt, Darmstadt, Germany, in 1989.

He holds the Chair for Microelectronic System Design in the Department of Electrical Engineering and Information Technology, University of Kaiserslautern–Landau, Kaiserslautern, Germany. He has more than 500 publications in various fields of microelectronic system design and holds several patents. His special research interests are VLSI-architectures for mobile communication, low-power techniques, advanced SoC and memory architectures, postquantum cryptography, reliability challenges in SoC, machine learning, IoT, and smart learning environments.



Georges Gielen (Fellow, IEEE) received the M.Sc. and Ph.D. degrees in electrical engineering from Katholieke Universiteit Leuven (KU Leuven), Leuven, Belgium, in 1986 and 1990, respectively.

He was a Postdoctoral Researcher with the University of California at, Berkeley, Berkeley, CA, USA. Since 1991, he has been with the MICAS Research Group, Department of Electrical Engineering, KU Leuven, where he is currently a Full Professor. His research interests include the design of analog and mixed-signal integrated circuits, such as sensor interfaces and data converters and analog and mixed-signal CAD tools and EDA.

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